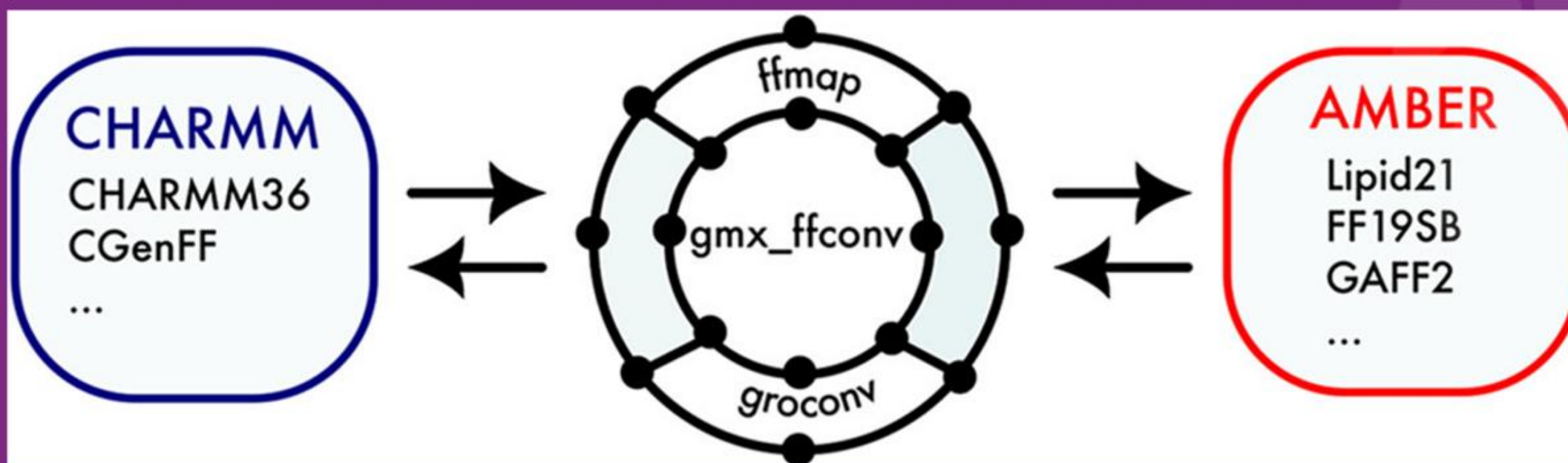




gmx_ffconv: Semi-Automated Force Field Conversion for GROMACS

Paper Information

- **Journal:** Journal of Chemical Information and Modeling
- **Year:** 2025
- **Topic:** GROMACS workflow automation / force field conversion / molecular graph matching
- **Links :** <https://doi.org/10.1021/acs.jcim.5c02200>



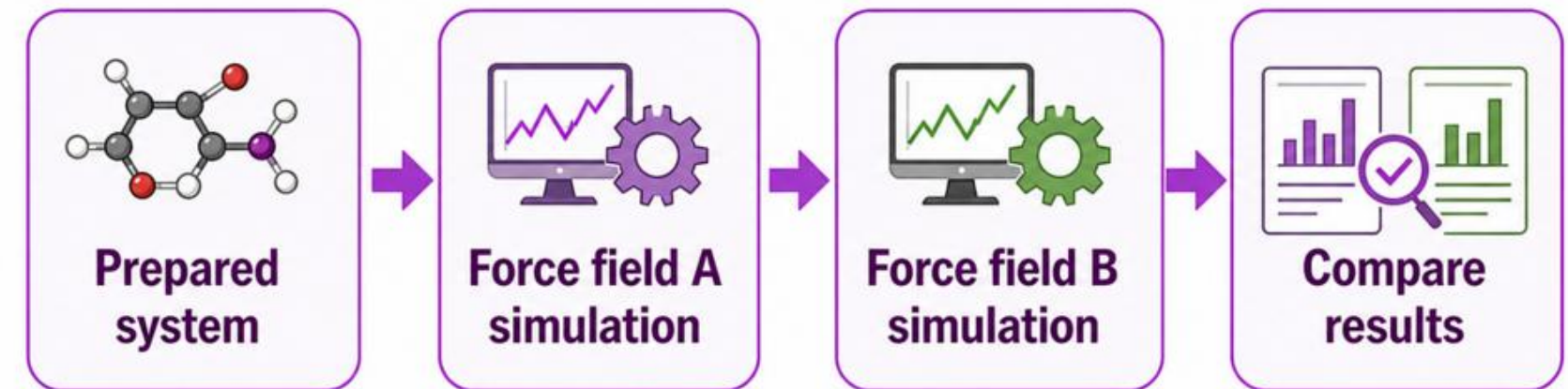
Presenter : Ayson HAN AIXIN

Date : 7/5/2026

Background: Why Force Field Conversion Matters

- MD simulation depends on **coordinates + topology**
- Different force fields are often used for:
 - validation
 - comparison
 - compatibility with existing workflows
- A fair comparison should start from **identical coordinates**.
- But switching force fields is not trivial

Comparative MD Simulations Using Different Force Fields

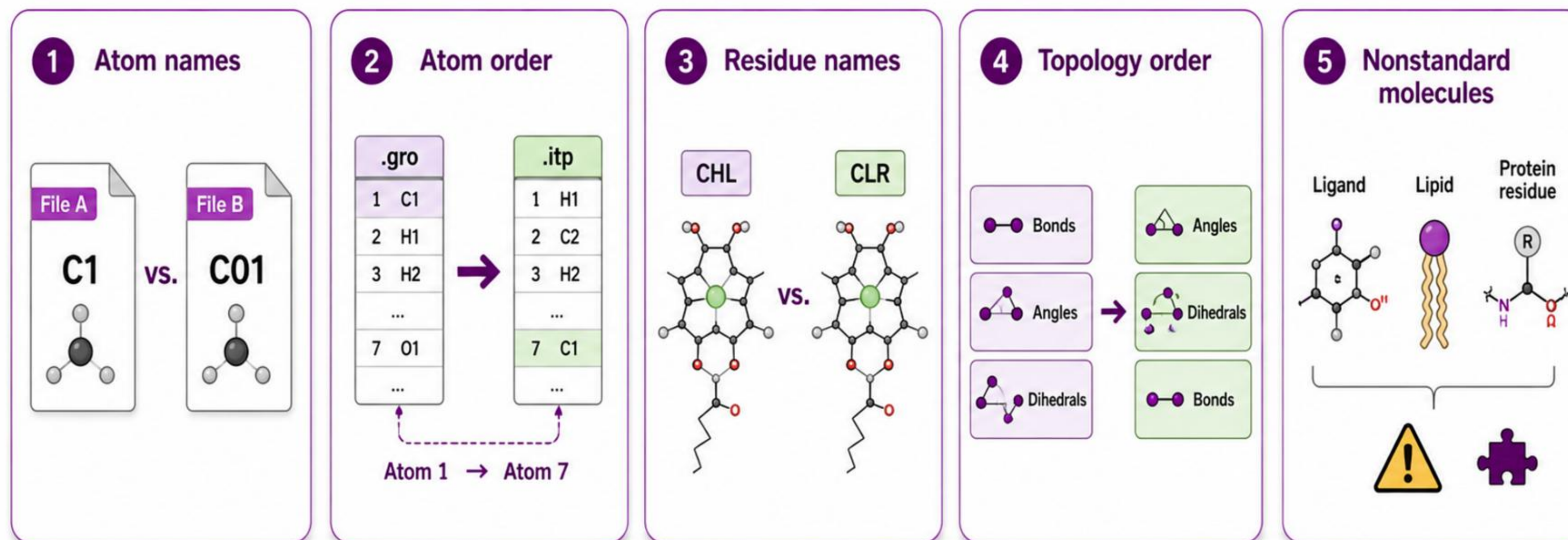


A fair comparison should begin from the same prepared system.

Why Is Force Field Conversion Difficult?

The main issue is mismatch between coordinates and topology when switching force fields.

Challenges in Force Field Conversion

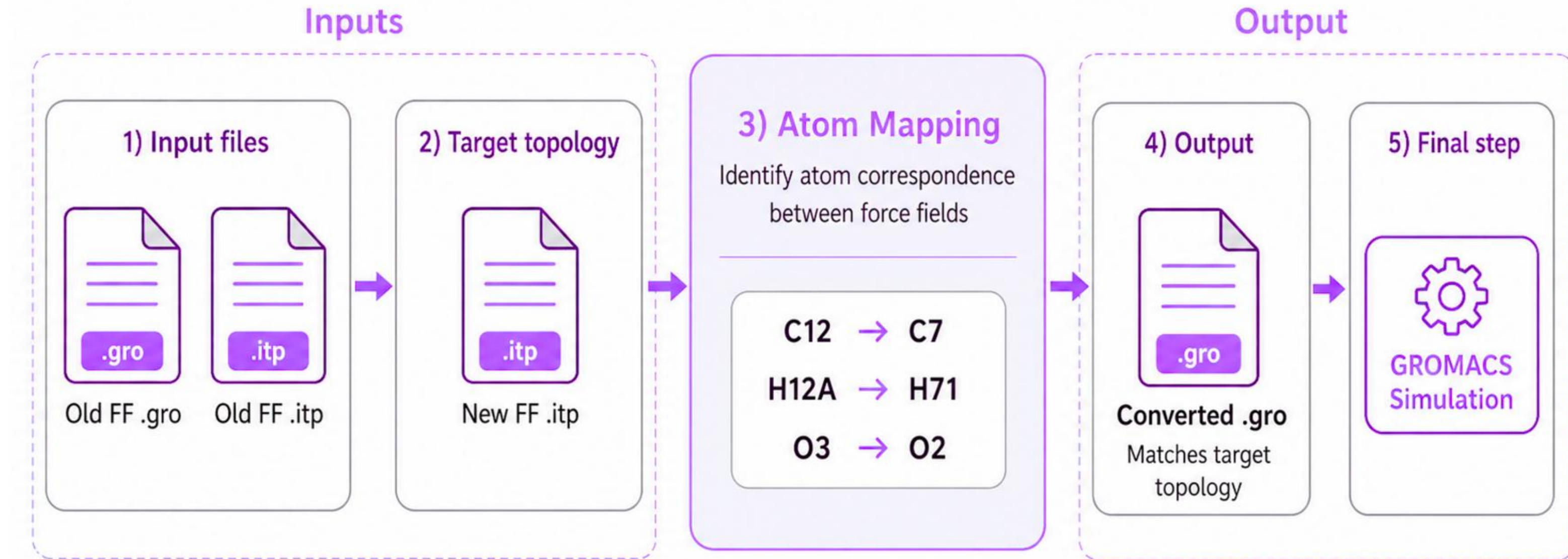


What Does gmx_ffconv Aim to Do?

What gmx_ffconv does ✓	What gmx_ffconv does NOT do ✗
Identify atom mapping	Generate new force field parameters
Reorder coordinates	Decide which force field is better
Support force field conversion in GROMACS	Fully automate all system preparation

- **Goal:** Convert a prepared GROMACS system between different all-atom force fields
- **Strategy:** Identify atom correspondence and reorder the coordinate file
- **Not designed to:** generate force field parameters

Overall Workflow of gmx_ffconv



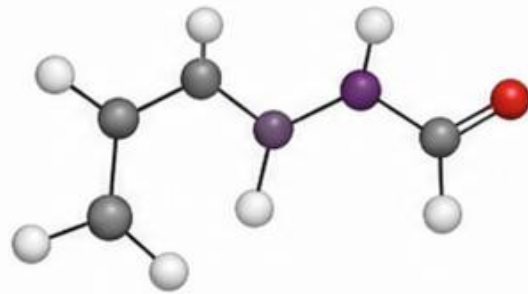
The coordinate file is reordered so that it is consistent with the target force field topology.

Core Method: Molecular Graph Matching

1. Molecular Representation

A molecule is represented as a graph:

- Atom = node
- Bond = edge



Examples:

small molecules, lipids, cholesterol, protein residues

2. Graph Comparison

Compare graphs



3. Atom Mapping and Conversion

Example mapping:

C12 → C7

H12A → H71

O3 → O2



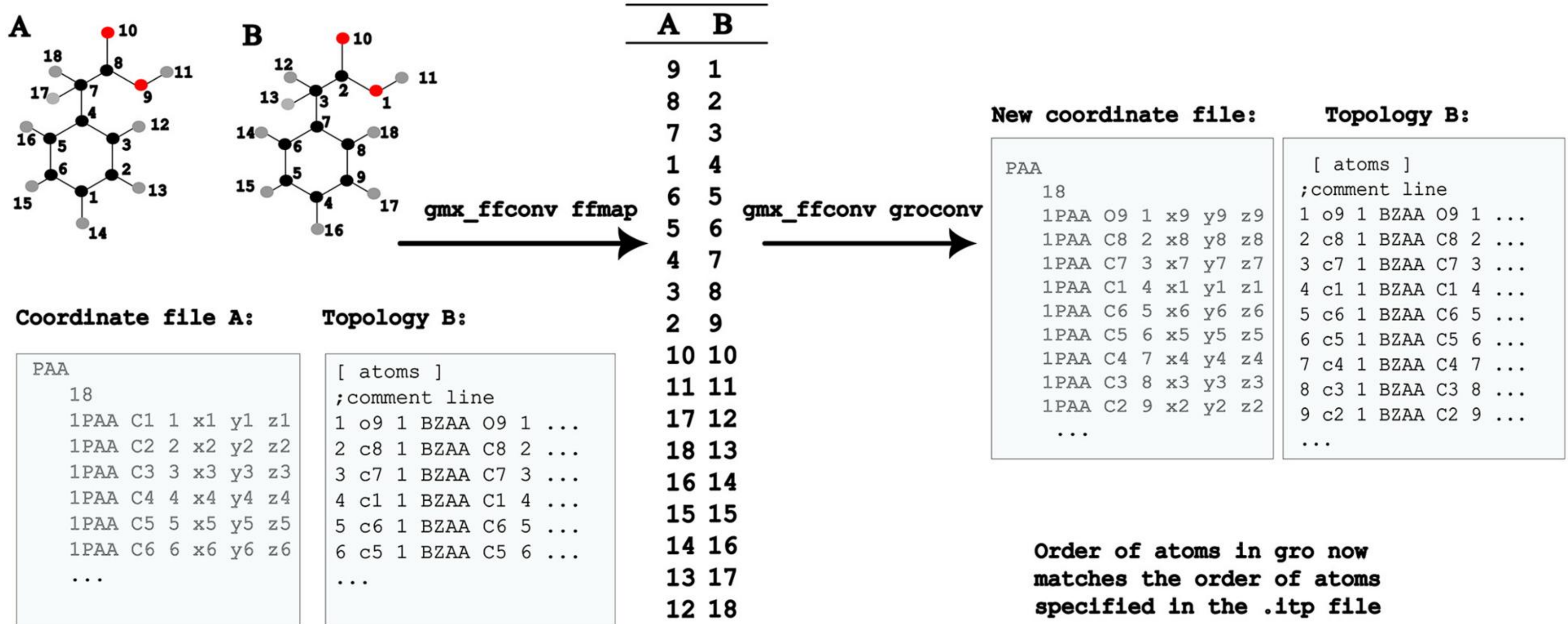
This mapping is used to reorder the .gro coordinate file so that it matches the target force field topology.



Result: converted coordinates are consistent with the new topology for GROMACS simulation.

The tool builds molecular graphs from two topology files and identifies atom correspondence based on graph connectivity.

Two-Step Procedure: ffdmap and groconv



Validation and Performance

Table 1 → Validation Scope **Table 2 → fffmap Timing** **Table 3 → Full-System Conversion**

System type	Scale
Small ligand	18 atoms
Viral membrane	2.27M atoms
HSA protein	18k atoms
Glycosylated spike	73k atoms

From small ligands to million-atom systems

Molecule	Time range
BZAA / CHL	~0.1 s
DPPC / DOPE	~60 s
HSA chains	~49 s

Mapping is usually fast, but molecule symmetry can slow it down.

System	ffmap	groconv
Phenylacetic acid	0.09 s	0.09 s
Viral membrane	208 s	4.47 s
Spike protein	1224 s	0.40 s

ffmap is the bottleneck; groconv is very fast.

Strengths and Limitations

Strengths	Limitations
reduces manual renaming and reordering	does not generate force field parameters
improves reproducibility	requires .itp files from both force fields
useful for comparative simulations	mainly designed for all-atom force fields
practical for large systems	converted systems still need validation

Take-Home Message and Relevance to MD Automation

- **Take-home message:**

Useful automation in MD does not replace scientific judgement.

Instead, it standardizes repetitive and error-prone preparation steps.

- **Relevance:**

This work is highly relevant to reproducible GROMACS workflow development.

